

S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

Seismo measures the magnitude of market-moving news

An AI/NLP risk engine that turns news, filings and social posts into calibrated, explainable risk signals, then tells a portfolio what each tremor costs.



Risk now moves at social-media speed; risk systems do not

\$42bn

withdrawn from Silicon Valley Bank on 9 March 2023, in a run organised in founders' group chats

\$100bn+

wiped off the Adani group in a week after a short seller's report in January 2023; RBI asked banks for their exposure

\$589bn

lost by Nvidia in one session on 27 January 2025, after a weekend of coverage of a cheaper AI model

1 fake

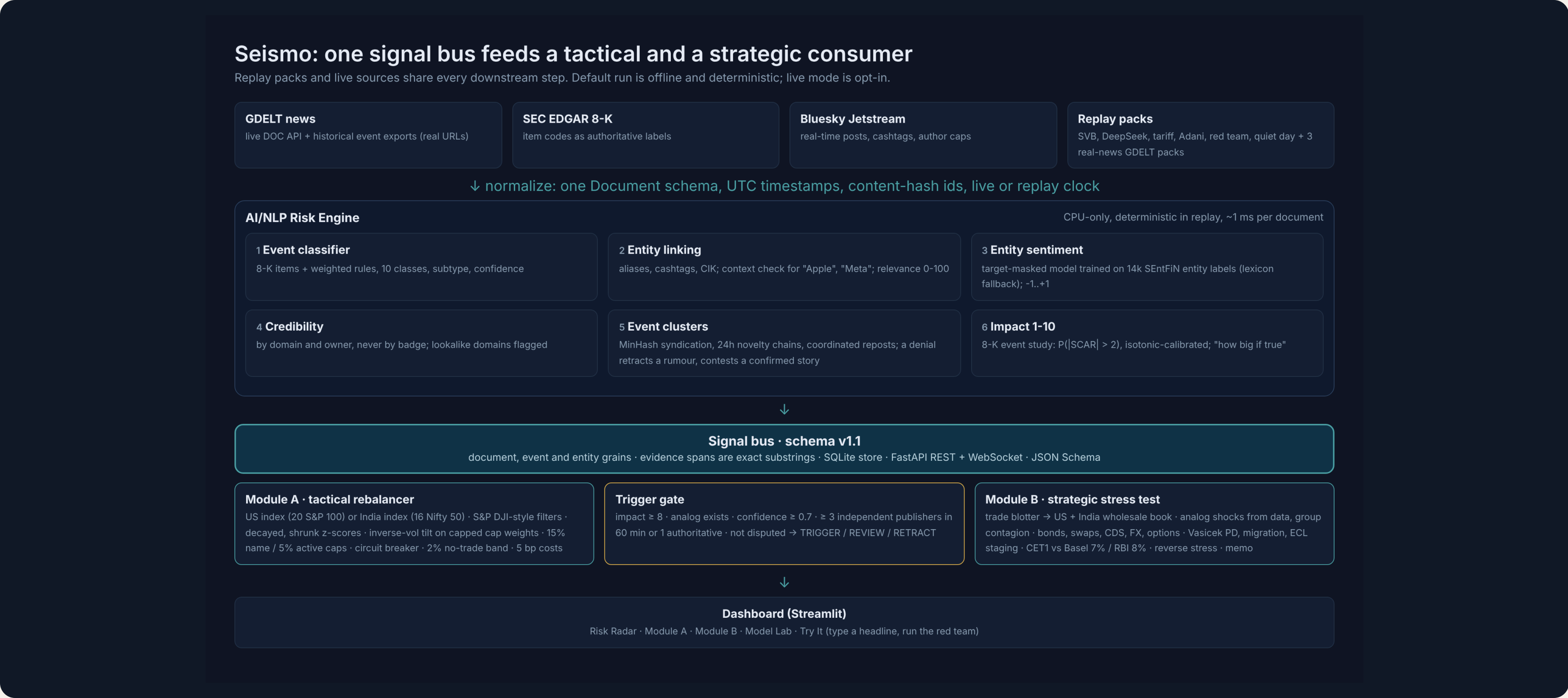
image of a Pentagon explosion, from an impersonating account, briefly moved the S&P 500 in May 2023

THE APPROACH

Score every story per company, calibrate its impact on real market reactions, let only corroborated events trigger action, and feed both a tactical index overlay and a bank stress test.

Sources: CNBC (SVB deposits); Bloomberg via DailyO and Business Standard (Adani); RBI press release, 3 Feb 2023; Bloomberg, Yahoo Finance (Nvidia); PetaPixel, Snopes (Pentagon image)

One signal bus feeds a tactical and a strategic consumer



Runs offline from replay packs with no API keys; real GDELT article streams, live SEC EDGAR and Bluesky adapters share the same pipeline and code paths.

Four choices make the signals trustworthy

Sentiment per company, learned from real labels

A target-masked model trained on 14,345 SEntFiN labels scores each company on its own: "Infosys gains while TCS slides" gives one up, one down. Linking checks context.

Events, not posts

Syndicated copies keep the original voice, sister outlets count once, and five or more near-identical accounts collapse into one coordinated voice. Novelty follows 24-hour chains.

Impact calibrated on real reactions

Impact is the chance of a two-sigma abnormal move, learned from 3,145 SEC 8-K events with a market-model event study. No language model sets the number.

A five-check trigger gate

Impact 8+, a historical analog, confidence 0.7+, three independent publishers or one authoritative source. A denial retracts a rumour but only contests a confirmed story.

Measured against naive baselines, on real data wherever it exists

RECONSTRUCTED CRISIS REPLAYS

Replay	Naive fires	Seismo	First trigger
SVB 2023	9	3	40 h before the closure
Adani 2023	7	3	14 h before the NSE open
DeepSeek 2025	6	1	3 h before Monday's open
Tariffs 2025	9	3	16 h before the first open
Red-team fake	1	0	held, then retracted
Quiet day	2	0	stayed calm

Naive: stress-test any document with $10 \times |\text{sentiment}|$ above 7. On real GDELT news Seismo also triggered on SVB and Adani, though only after GDELT first carried each story (a rule fix after an SVB miss is disclosed in the repo).

0.82 vs 0.61

sentiment accuracy per company on 2,869 held-out SEntFiN pairs (real Economic Times headlines), vs a finance lexicon; 0.69 vs 0.46 when two companies move in opposite directions

0 vs 48

stress tests run on 816 real articles in a calm week (GDELT, May 2024): Seismo vs the naive rule

0.77 vs 0.73

impact AUC for a two-sigma move on 2022–23 SEC 8-K events, vs a hand-set severity ranking

Every number regenerates with make sentiment and make results. Replays are labelled synthetic reconstructions; SEntFiN, GDELT, prices, rates and 8-K filings are real public data.

For an Indian bank, a headline about one group becomes a capital number



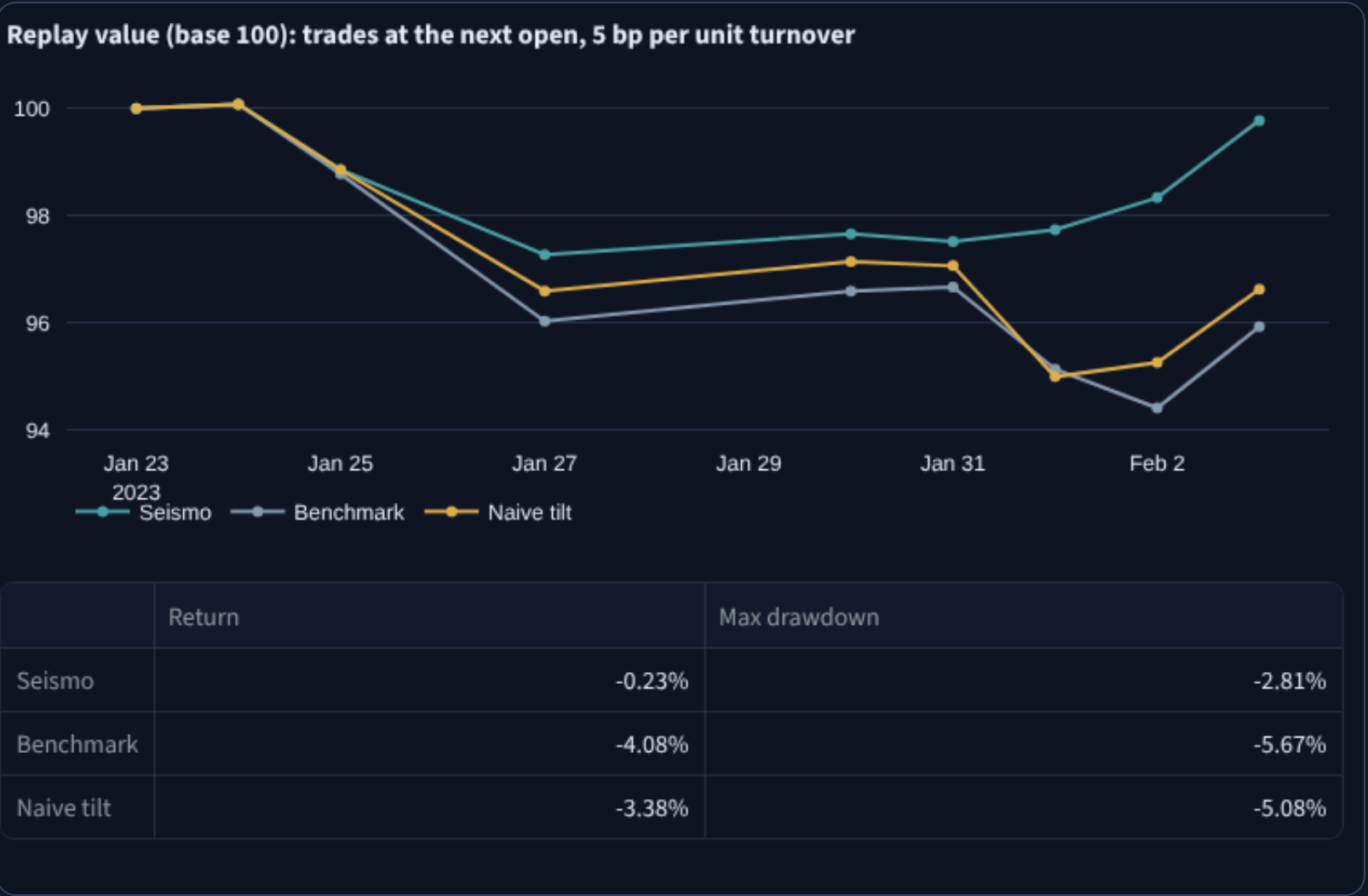
ADANI-HINDENBURG REPLAY, REAL NSE DATA

CET1 13.00% → 12.67%

Only Indian factors move: Adani Enterprises –54.5%, Adani Ports –39.2%, Nifty Bank –5.6%. Adani Ports takes two-thirds of the downgrade (group contagion); no US industrial is touched. The RBI 8% floor breaks only at 6.6× the shock, matching RBI's 3 February 2023 view that banks stayed resilient.

Module A cut Adani Enterprises from 5.2% to 0.2% of a 16-stock Nifty index 15 hours before the open: –0.2% over the window vs –4.1% for the benchmark.

Built for RBI's expected-credit-loss directions (effective 1 April 2027): three-stage ECL, CET1 against Basel 7% and RBI 8%, every number traceable to its evidence.



Synthetic wholesale book (257 positions; Adani lines about 4%). Shocks are each factor's largest move in the crisis window, computed from Yahoo Finance and FRED data.

What it does not do yet, and what comes next

LIMITS, STATED PLAINLY

- The event classifier is rule-based and the weakest part on real headlines; it missed real SVB news until bank-crisis words were added
- Archived social posts are not free, so crisis replays are labelled reconstructions; GDELT headlines are rebuilt from URLs
- Sentiment 0.82 vs about 0.93 for GPU transformers in the SEntFiN paper
- Today's index membership and share counts: survivorship bias

NEXT MONTH

- A learned event classifier, scored on the same real-news windows
- A fine-tuned transformer with the same target masking
- Hindi and regional-language news for the India view
- Supply-chain and counterparty spillovers; point-in-time constituents